

HARSH MAURYA

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EDUCATION

Manipal University Jaipur

Jaipur, Rajasthan, India

Bachelor of Technology, Computer Science Engineering — Specialization: Data Science

Expected May 2028

CGPA: 8.64/10.0 **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Statistical Modeling, Object-Oriented Programming, Probability & Statistics

TECHNICAL SKILLS

Programming Languages: Python, C++, Java

ML & AI Frameworks: TensorFlow, PyTorch, Scikit-learn, Keras, Hugging Face Transformers

ML Concepts & Methods: Supervised Learning, Unsupervised Learning, Reinforcement Learning, Neural Networks, CNNs, RNNs, Transformers, Attention Mechanisms, Transfer Learning, Model Fine-Tuning, Hyperparameter Optimization, Feature Engineering, A/B Testing

NLP & LLMs: Large Language Models (LLMs), LLM Agents, Prompt Engineering, Text Classification, Tokenization, Speech-to-Text, Text-to-Speech, Conversational AI

Computer Vision: OpenCV, MediaPipe, Image Classification, Object Detection, Gesture Recognition, Real-Time Inference

Data Science & Analysis: Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Statistical Modeling, Data Visualization

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, Linux, VS Code

PROJECTS

Agentic AI Medical Assistant System

Python · LLM Agents · NLP · Voice AI · Prompt Engineering

- Architected a full-stack **agentic AI** healthcare platform where users describe symptoms in natural language; an **LLM agent** analyzes the free-text input and surfaces a curated, ranked list of relevant specialist doctors using **prompt engineering** and structured output generation.
- Engineered an intelligent **voice chat** interface enabling users to converse with an AI-simulated doctor in real time — delivering personalized medication guidance, treatment recommendations, and medical assistance through **speech-to-text** and **text-to-speech** pipelines.
- Built an interactive **medical report dashboard** that aggregates AI-generated consultation summaries, prescribed medications, and diagnostic insights into a structured, user-readable format for post-session review.
- Implemented a **patient history module** storing past consultation sessions and symptom records, enabling the LLM agent to incorporate longitudinal context for more accurate, **context-aware medical reasoning** across multiple interactions.

Real-Time Hand Sign Translator

Python · OpenCV · MediaPipe · CNN · Scikit-learn

- Built a **real-time computer vision** pipeline capturing live hand gestures via webcam, processing landmark coordinates using **MediaPipe** hand tracking, and classifying gestures using a trained **machine learning** model with sub-100ms inference latency.
- Conducted end-to-end **feature engineering** on 21-point hand landmark data — normalizing, scaling, and constructing relative-coordinate features — to improve model generalization across users and lighting conditions.
- Trained and evaluated multiple classifiers (**SVM**, **Random Forest**, **CNN**) using **cross-validation** and **hyperparameter tuning**, selecting the optimal model based on F1-score and inference speed trade-offs.
- Architected an **extensible modular system** supporting dynamic gesture sequences and full **sign language recognition**, enabling future integration with assistive technology and accessibility applications.

Movie Recommendation System

Python · Scikit-learn · Pandas · NumPy · Collaborative Filtering

- Developed a **hybrid recommendation engine** combining **collaborative filtering** (matrix factorization, cosine similarity) and **content-based filtering** (TF-IDF feature vectors) to deliver personalized movie recommendations.
- Performed large-scale **exploratory data analysis (EDA)** and **data preprocessing** on a movie ratings dataset — handling missing values, engineering interaction features, and removing data sparsity via threshold-based filtering.
- Benchmarked model variants using **RMSE**, **Precision@K**, and **Recall@K** metrics, conducting **A/B comparison** across filtering strategies to identify the highest-performing recommendation algorithm.

ACHIEVEMENTS & ACTIVITIES

- Maintaining **CGPA of 8.64/10.0** in B.Tech CSE (Data Science) at Manipal University Jaipur — demonstrating consistent academic excellence across mathematics, ML theory, and systems courses.
- Actively building expertise in **transformer architectures**, **LLM fine-tuning**, and **agentic AI** systems through self-directed research and project implementation.
- Engaged with open-source ML community on **GitHub** — documenting projects with reproducible pipelines, model cards, and evaluation benchmarks.